

2024, Mar. 2024 Release LOC:Acute Adult

Pancreatitis

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ^(1, 2, 3, 4)Episode Day 1 ^(1, 3, 4)Episode Day 2 ^(3, 4, 48)Episode Day 3 ^(4, 83, 84)

Episode Day 4

Discharge Screens ⁽⁸⁹⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is intended for patients with acute pancreatitis or those with an acute exacerbation of chronic pancreatitis that require medical management. For chronic pancreatitis requiring surgical intervention, use the General Surgical subset. For biliary sepsis, use the Infection: GI/GYN subset.

Introduction:

Acute pancreatitis is defined as inflammation of the pancreas which may extend to the surrounding tissue. The most common cause of acute pancreatitis is gallstones, followed by alcohol ingestion and hypertriglyceridemia. Acute pancreatitis may range from mild to severe as determined by the Atlanta Classification (Banks et al., Gut 2013, 62: 102-11). Determining severity is important to ensure care is delivered in the appropriate setting and to govern optimal therapy. However, clinical scoring systems to accurately reflect severity are difficult to use, especially in the first 24 hours of admission (Waller et al., J Emerg Med 2018, 55: 769-79).

Mild disease lacks significant systemic involvement and follows an uncomplicated, self-limiting course that usually resolves within a week (Janisch and Gardner, Gastroenterol Clin North Am 2016, 45: 1-8). Severe disease occurs in up to 20% of cases and involves local or systemic complications which can rapidly progress to multisystem organ failure and death. Local complications may include pancreatic necrosis, abscess formation, and pseudocysts peripancreatic fluid collections. Severe acute pancreatitis is delineated into two phases, early or late. Early disease typically resolves within 1-2 weeks and is characterized by systemic inflammatory response (SIRS) and/or organ failure. Late disease occurs after 1 week and can last months, it is associated with local complications as described previously (Crockett et al., Gastroenterology 2018, 154: 1096-101; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16).

Chronic pancreatitis is a condition in which episodes of pancreatic inflammation lead to fibrosis of pancreatic tissue that may cause chronic pain, exocrine and endocrine pancreatic insufficiency, poor quality of life, and shorter life expectancy. A new mechanistic definition of chronic pancreatitis determines earlier diagnosis and prediction of the clinical course, as well as possible identification of preventative treatments along with symptomatic and supportive care (Gardner et al., Am J Gastroenterol 2020, 115: 322-39; Whitcomb et al., Pancreatol 2016, 16: 218-24). Acute and recurrent acute pancreatitis episodes are believed to progress to chronic pancreatitis overtime based on an individual's clinical presentation, complications, and risk factors. Approximately 50% of patients with chronic pancreatitis will continue to have recurrent episodes of acute pancreatitis (Singh et al., JAMA 2019, 322: 2422-34). Patients with chronic pancreatitis have a chronic course associated with development of endocrine and nutritional complications, such as diabetes and pancreatic exocrine insufficiency (Sharma et al., Cureus 2021, 13: e14242; Beyer et al., Lancet 2020, 396: 499-512). The mortality rate in patients with chronic pancreatitis is lower than patients with severe acute pancreatitis due to a proposed cushion of fibrosis that surrounds the pancreas preventing the spread of new-onset necrosis. Risk

factors for chronic pancreatitis are multifactorial including genetic, environmental and/or other, such as excessive alcohol use or smoking.

Evaluation and Treatment:

The diagnosis of acute pancreatitis requires two of the following three features: (1) abdominal pain consistent with acute pancreatitis, (2) serum lipase activity (or amylase activity) at least three times greater than the upper limit of normal; and (3) characteristic findings of acute pancreatitis on imaging (Banks et al., Gut 2013, 62: 102-11). Currently, there is no gold-standard for the diagnosis of chronic pancreatitis and a combination of the following are used: exposure risk, underlying predisposition, cross-sectional imaging, and direct and/or indirect pancreatic function tests (Gardner et al., Am J Gastroenterol 2020, 115: 322-39). The mechanistic definition considers the individual's clinical features, risk factors, and biomarkers (imaging and direct/indirect pancreatic function test) in combination to make a diagnosis (Whitcomb et al., Pancreatology 2016, 16: 218-24). While the diagnosis remains a clinical diagnosis, histologic confirmation may also be valuable in ruling out chronic pancreatitis.

Supportive care is the mainstay of treatment in hospitalized patients with pancreatitis. Specifically, care should be directed towards maintaining oxygenation, goal directed fluid therapy, pain management, nutritional support, and preventing complications. Other treatment options for patients with pancreatitis who fail medical management may include surgery, drainage of fluid collections, or antibiotic use (Sharma et al., Cureus 2021, 13: e14242). Anti-infective therapy may be appropriate when sepsis or infected pancreatic necrosis is suspected or confirmed. The routine use of anti-infective prophylaxis is not recommended at this time (Crockett et al., Gastroenterology 2018, 154: 1096-101; Greenberg et al., Can J Surg 2016, 59: 128-40; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16). Emerging treatments such as insulin, heparin, fibric acids, and omega 3 fatty acids have been used to reduce serum triglycerides in patients with hypertriglyceride-induced pancreatitis. However, as no approved guidelines for the treatment of patients with hypertriglyceride-induced pancreatitis currently exist, treatment should follow current guidelines for acute pancreatitis (Rawla et al., Clin J Gastroenterol 2018, 11: 441-8).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾● Pancreatitis actual or suspected, ≥ **Two:**

- Abdominal pain consistent with pancreatitis ⁽⁶⁾
- Amylase or lipase ≥ 3x ULN ⁽⁷⁾
- Pancreatitis confirmed by imaging ^(8, 9, 10, 11)

● **ACUTE, One:** ⁽¹²⁾● Pancreatitis and high risk for severe disease, **Both:**● Finding, ≥ **Two:**

- Abdominal pain consistent with pancreatitis ⁽⁶⁾
- Amylase or lipase ≥ 3x ULN ⁽⁷⁾
- Pancreatitis confirmed by imaging ^(8, 9, 10, 11)

● Risk factor, ≥ **One:** ⁽¹³⁾

- Hct ≥ 44%(0.44) after aggressive fluid resuscitation
- Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹⁴⁾
- Pancreatic duct obstruction ⁽¹⁵⁾
- Pleural effusion confirmed by imaging ⁽¹⁶⁾
- Pulmonary infiltrate confirmed by imaging ⁽¹⁷⁾

● **INTERMEDIATE, One:** ⁽¹⁸⁾● Pancreatitis, **Both:**● Finding, ≥ **Two:**

- Abdominal pain consistent with pancreatitis ⁽⁶⁾
- Amylase or lipase ≥ 3x ULN ⁽⁷⁾
- Pancreatitis confirmed by imaging ^(8, 9, 10, 11)

● Complication, ≥ **One:**● Respiratory insufficiency, ≥ **One:**

- Arterial or venous Pco₂ > 54 mmHg(7.2 kPa) and pH < 7.31 ⁽¹⁹⁾
- O₂ sat < 89%(0.89) and < baseline or room air assessment not clinically appropriate ^(20, 21)

● SIRS, ≥ **Two:** ^(22, 23)

- Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽²⁴⁾

● WBC, ≥ **One:**

- > 12,000/cu.mm(12x10⁹/L) or < 4,000/cu.mm(4x10⁹/L)
- Bands > 10%(0.10)

- Heart rate > 90/min, sustained ⁽²⁵⁾

- Respiratory rate ≥ 20/min, sustained ⁽²⁵⁾

● **CRITICAL, One:** ^(26, 27, 28)● Pancreatitis and, ≥ **One:**

- Calcium < 5.0 mg/dL(1.25 mmol/L) ⁽²⁹⁾
- DIC and blood product transfusion ⁽³⁰⁾
- ECMO or ECLS ⁽³¹⁾

● Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** ⁽³²⁾

- MAP < 65 mmHg
- Systolic blood pressure < 90 mmHg and < baseline
- Hemodynamic monitoring, invasive ⁽³³⁾
- Mechanical ventilation

● NIPPV and, ≥ **One:** ⁽³⁴⁾

- Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
- Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**

- COPD and arterial or venous pH ≤ 7.24
- No hx of COPD
- Sodium > 160 mEq/L(160 mmol/L) ⁽³⁵⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 1, One:** ^(1, 3, 4)
(Symptom or finding within 24h)
- **OBSERVATION, One:** ⁽⁵⁾
 - Pancreatitis, actual or suspected, **Both:**
 - Finding, $\geq$ **Two:**
 - Abdominal pain consistent with pancreatitis ⁽⁶⁾
 - Amylase or lipase $\geq 3x$ ULN ⁽⁷⁾
 - Pancreatitis confirmed by imaging ^(8, 9, 10, 11)
 - Intervention, **Both:** ⁽³⁶⁾
 - Acute pain and failed outpatient analgesic and, **Both:** ⁽³⁷⁾
 - Analgesic 2-3x/24h
 - Unresponsive to ≥ 2 doses of analgesic in ED prior to admit decision
 - IV fluid resuscitation $\leq 2d$ ^(38, 39)
- **ACUTE, One:** ⁽¹²⁾
 - Pancreatitis and high risk for severe disease, **All:**
 - Finding, $\geq$ **Two:**
 - Abdominal pain consistent with pancreatitis ⁽⁶⁾
 - Amylase or lipase $\geq 3x$ ULN ⁽⁷⁾
 - Pancreatitis confirmed by imaging ^(8, 9, 10, 11)
 - Risk factor, $\geq$ **One:** ⁽¹³⁾
 - Hct $\geq 44\%(0.44)$ after aggressive fluid resuscitation
 - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹⁴⁾
 - Pancreatic duct obstruction ⁽¹⁵⁾
 - Pleural effusion confirmed by imaging ⁽¹⁶⁾
 - Pulmonary infiltrate confirmed by imaging ⁽¹⁷⁾
 - Intervention, **Both:**
 - Acute pain and, $\geq$ **One:** ⁽³⁷⁾
 - Analgesic, continuous
 - Analgesic $\geq 4x/24h$
 - IV fluid resuscitation $\leq 2d$ ^(38, 39)
- **INTERMEDIATE, One:** ⁽¹⁸⁾
 - Pancreatitis, **Both:**
 - Finding, $\geq$ **Two:**
 - Abdominal pain consistent with pancreatitis ⁽⁶⁾
 - Amylase or lipase $\geq 3x$ ULN ⁽⁷⁾
 - Pancreatitis confirmed by imaging ^(8, 9, 10, 11)
 - Complication, $\geq$ **One:**
 - Respiratory insufficiency, **Both:**
 - Finding, $\geq$ **One:**
 - Arterial or venous Pco₂ > 54 mmHg(7.2 kPa) and pH < 7.31 ⁽¹⁹⁾
 - O₂ sat $< 89\%(0.89)$ and $<$ baseline or room air assessment not clinically appropriate ^(20, 21)
 - Intervention, **Both:**
 - Respiratory, $\geq$ **One:**
 - NIPPV ⁽³⁴⁾
 - Oxygen $\geq 40\%(0.40)$ ⁽⁴⁰⁾
 - IV fluid resuscitation $\leq 2d$ ⁽³⁹⁾
 - SIRS, **Both:** ^(22, 23)
 - Finding, $\geq$ **Two:**
 - Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$ or $< 97.0^{\circ}\text{F}(36.1^{\circ}\text{C})$ ⁽²⁴⁾

- **WBC, $\geq$ One:**
 - $> 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$ or $< 4,000/\text{cu.mm}(4 \times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - Heart rate $> 90/\text{min}$, sustained ⁽²⁵⁾
 - Respiratory rate $> 20/\text{min}$, sustained ⁽²⁵⁾
 - Volume expander $\leq 2\text{d}$ ⁽⁴¹⁾
- **CRITICAL, $\geq$ One:** ^(26, 27, 28)
 - Biliary sepsis (cholangitis) (**use Infection GI/GYN subset**) ^(42, 43, 44)
- **Pancreatitis and, $\geq$ One:**
 - DIC and blood product transfusion ⁽³⁰⁾
 - ECMO or ECLS ⁽³¹⁾
- **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24\text{h}$ and, $\geq$ One:** ⁽³²⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Hemodynamic monitoring, invasive ⁽³³⁾
- **Hypocalcemia, Both:** ^(29, 45)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Mechanical ventilation
- **NIPPV and, $\geq$ One:** ⁽³⁴⁾
 - Arterial $\text{Po}_2 \leq 39$ mmHg(5.2 kPa)
 - **Arterial or venous $\text{Pco}_2 \geq 60$ mmHg(8.0 kPa) and, One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Sodium > 160 mEq/L(160 mmol/L) and IV fluid resuscitation ^(35, 46, 47)
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:** ^(3, 4, 48)
 - **OBSERVATION, One:** ⁽⁵⁾
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁹⁾
 - Pancreatitis resolving
 - Pain controlled or manageable
 - Tolerating PO ⁽⁵⁰⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - **Pancreatitis unresolved, Both:** ⁽³⁶⁾
 - Acute pain and analgesic 2-3x/24h ⁽³⁷⁾
 - IV fluid resuscitation $\leq 2\text{d}$ ^(38, 39)
- **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁹⁾
 - Pancreatitis resolving
 - Pain controlled or manageable
 - Tolerating PO or nutritional route established ^(50, 51)
 - No complication or active comorbidity relevant to this episode of care ⁽⁵²⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Cholecystectomy required (**use General Surgical subset**)
 - **Pancreatitis and high risk for severe disease, Both:**
 - **Acute pain and, $\geq$ One:** ⁽³⁷⁾
 - Analgesic, continuous
 - Analgesic $\geq 4\text{x}/24\text{h}$
 - IV fluid resuscitation $\leq 2\text{d}$ ^(38, 39)
 - **AKI, Both:** ⁽⁵³⁾
 - **Finding, $\geq$ One:**
 - Oliguria ^(54, 55)

- **Creatinine, One:**
 - $\geq 1.5\times$ baseline and chronic kidney disease
 - $\geq 1.5\times$ ULN ⁽⁵⁶⁾
 - GFR $> 25\%(0.25)$ decrease from baseline ⁽²⁰⁾
 - Dialysis initiated
- **Intervention, $\geq$ One:**
 - Volume expander $\leq 2d$ ^(41, 57)
 - Diuretic $\geq 2\times/24h, \leq 3d$
 - Dialysis (initial) $\leq 7d$ since initiation ^(58, 59)
 - Dialysis discontinued requiring lab monitoring $\leq 2d$ since discontinuation ⁽⁶⁰⁾
 - Immunosuppressant therapy $\leq 3d$ ^(61, 62, 63)
- ERCP $\leq 24h$ ^(64, 65)
- **Blood glucose $\leq 24h$, All:** ⁽⁶⁶⁾
 - **Finding, $\geq$ One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(66, 67)
- **DVT confirmed by ultrasound and, One:** ^(68, 69, 70)
 - Anticoagulation (includes PO) $\leq 24h$ ⁽⁷¹⁾
 - Unfractionated heparin, continuous $\leq 5d$ ⁽⁷²⁾
- **Ileus and, One:** ⁽⁷³⁾
 - Resolved and initiating PO or tube feeding $\leq 2d$
 - Unresolved and IV fluid $\leq 5d$ ⁽⁷⁴⁾
- **Inadequate oral intake and, $\geq$ One:** ⁽⁷⁵⁾
 - Initiating oral or tube feeding $\leq 2d$ ^(76, 77)
 - IV fluid ⁽⁷⁴⁾
- O₂ sat $\leq 91\%(0.91)$ and $<$ baseline requiring supplemental oxygen ^(20, 40)
- Post Critical care $\leq 24h$
- **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽³⁴⁾
 - Oxygen $\geq 40\%(0.40), \leq 2d$ ⁽⁴⁰⁾
- **CRITICAL, $\geq$ One:**
 - **AKI and, Both:** ⁽⁵³⁾
 - **Finding, $\geq$ One:**
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁷⁸⁾
 - Widening QRS
 - **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - **Intervention, $\geq$ One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁹⁾
 - Diuretic, continuous
 - **Dialysis, One:**
 - Hemodialysis $\leq 7d$ since initiation
 - Peritoneal dialysis $\leq 7d$ since initiation
 - IV fluid resuscitation $\leq 2d$ ⁽⁸⁰⁾
 - CRRT ⁽⁸¹⁾

- DIC and blood product transfusion ⁽³⁰⁾
- ECMO or ECLS ⁽³¹⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, $\geq$ **One:** ⁽³²⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
- Hemodynamic monitoring, invasive ⁽³³⁾
- Hypocalcemia, **Both:** ^(29, 45)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽³⁴⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽⁸²⁾
- Sodium > 160 mEq/L(160 mmol/L) and IV fluid resuscitation ^(35, 47)
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One:** ^(4, 83, 84)
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁹⁾
 - Pancreatitis resolving
 - Pain controlled or manageable
 - Tolerating PO ⁽⁵⁰⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For pancreatitis use Episode Day 3 at a higher level of care
 - For a condition other than pancreatitis use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁹⁾
 - Pancreatitis resolving
 - Pain controlled or manageable
 - Tolerating PO or nutritional route established ^(50, 51)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁵²⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:**
 - **Acute kidney injury (AKI), $\geq$ One:**
 - Kidney function improving or within acceptable limits ⁽⁸⁵⁾
 - Outpatient dialysis regimen established
 - Biliary stones or obstruction resolved
 - Blood glucose within acceptable limits ⁽⁸⁵⁾
 - Anticoagulation regimen established and anticoagulation therapeutic ^(86, 87)
 - O₂ sat within acceptable limits ⁽⁸⁵⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Cholecystectomy required (**use General Surgical subset**)
 - **Pancreatitis and high risk for severe disease and, Both:**
 - **Acute pain and, $\geq$ One:** ⁽³⁷⁾
 - Analgesic, continuous
 - Analgesic ≥ 4 x/24h
 - **Hydration or nutrition, $\geq$ One:**
 - Initiating PO or tube feeding ≤ 2 d ^(76, 77)
 - IV fluid ⁽⁷⁴⁾
 - **AKI, Both:** ⁽⁵³⁾
 - **Finding, $\geq$ One:**

- Oliguria ^(54, 55)
- **Creatinine, One:**
 - $\geq 1.5\times$ baseline and chronic kidney disease
 - $\geq 1.5\times$ ULN ⁽⁵⁶⁾
 - GFR $> 25\%(0.25)$ decrease from baseline ⁽²⁰⁾
 - Dialysis initiated
- **Intervention, $\geq$ One:**
 - Volume expander $\leq 2d$ ^(41, 57)
 - Diuretic $\geq 2\times/24h, \leq 3d$
 - Dialysis (initial) $\leq 7d$ since initiation ^(58, 59)
 - Dialysis discontinued requiring lab monitoring $\leq 2d$ since discontinuation ⁽⁶⁰⁾
 - Immunosuppressant therapy $\leq 3d$ ^(61, 62, 63)
- ERCP $\leq 24h$ ^(65, 88)
- **Blood glucose $\leq 24h$, All:** ⁽⁶⁶⁾
 - **Finding, $\geq$ One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(66, 67)
- **DVT confirmed by ultrasound and, One:** ^(69, 70)
 - Anticoagulation (includes PO) $\leq 24h$ ⁽⁷¹⁾
 - Unfractionated heparin, continuous $\leq 5d$ ⁽⁷²⁾
- **Ileus and, One:** ⁽⁷³⁾
 - Resolved and initiating PO or tube feeding $\leq 2d$
 - Unresolved and IV fluid $\leq 5d$ ⁽⁷⁴⁾
- **Inadequate oral intake and, $\geq$ One:** ⁽⁷⁵⁾
 - Initiating oral or tube feeding $\leq 2d$ ^(76, 77)
 - IV fluid ⁽⁷⁴⁾
- O_2 sat $\leq 91\%(0.91)$ and $<$ baseline requiring supplemental oxygen ^(20, 40)
- Post Critical care $\leq 24h$
- **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽³⁴⁾
 - Oxygen $\geq 40\%(0.40) \leq 2d$ ⁽⁴⁰⁾
- **CRITICAL, $\geq$ One:** ⁽²⁷⁾
 - **AKI and, Both:** ⁽⁵³⁾
 - **Finding, $\geq$ One:**
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁷⁸⁾
 - Widening QRS
 - **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - **Intervention, $\geq$ One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁹⁾
 - Diuretic, continuous
 - **Dialysis, One:**
 - Hemodialysis $\leq 7d$ since initiation
 - Peritoneal dialysis $\leq 7d$ since initiation
 - IV fluid resuscitation $\leq 2d$ ⁽⁸⁰⁾

- CRRT ⁽⁸¹⁾
- DIC and blood product transfusion ⁽³⁰⁾
- ECMO or ECLS ⁽³¹⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, $\geq$ **One:** ⁽³²⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
- Hemodynamic monitoring, invasive ⁽³³⁾
- Hypocalcemia, **Both:** ^(29, 45)
 - Calcium < 5.0 mg/dL(1.25 mmol/L)
 - Calcium repletion
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽³⁴⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽⁸²⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 4, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁹⁾
 - Pancreatitis resolving
 - Pain controlled or manageable
 - Tolerating PO or nutritional route established ^(50, 51)
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁵²⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:**
 - **Acute kidney injury (AKI), $\geq$ One:**
 - Kidney function improving or within acceptable limits ⁽⁸⁵⁾
 - Outpatient dialysis regimen established
 - Biliary stones or obstruction resolved
 - Blood glucose within acceptable limits ⁽⁸⁵⁾
 - Anticoagulation regimen established and anticoagulation therapeutic ^(86, 87)
 - O₂ sat within acceptable limits ⁽⁸⁵⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One:** ⁽²⁷⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁸⁹⁾

- **HOME, Both:**
 - **Level of care appropriateness, Both:**
 - Home environment safe and accessible ⁽⁹⁰⁾
 - Patient or caregiver demonstrates ability to manage care
- **Pancreatitis discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions ⁽⁹¹⁾
- Education on the importance of, **All:**
 - Adhering to follow-up appointments
 - Abstaining from alcohol
 - Maintaining a low-fat diet
 - Seeking medical attention when appropriate
- Medication reconciliation ⁽⁹²⁾
- Follow-up care planned ⁽⁹³⁾
- Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁹⁰⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(94, 95)
 - Skilled services, ≥ **One:** ⁽⁹⁶⁾
 - Skilled nursing ⁽⁹⁷⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁹⁸⁾
 - Pancreatitis discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁹¹⁾
 - Education on the importance of, **All:**
 - Adhering to follow-up appointments
 - Abstaining from alcohol
 - Maintaining a low-fat diet
 - Seeking medical attention when appropriate
 - Medication reconciliation ⁽⁹²⁾
 - Follow-up care planned ⁽⁹³⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁹⁹⁾
 - Skilled treatment, ≥ **One:**
 - Clinical assessment ⁽¹⁰⁰⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽¹⁰¹⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁹⁷⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(102, 103)
 - Functional impairments requiring at least supervision ⁽¹⁰⁴⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁹²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk

- Treatment precluded in a lower level
- **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁹⁷⁾
- **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(102, 103)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁰⁴⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁹²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(102, 103)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁰⁴⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁹²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - **Level of care appropriateness, Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(102, 103)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁰⁴⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽¹⁰⁵⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁹²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
 - **In lieu of acute or continued hospitalization or failed lower level of care, ≥ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽¹⁰⁶⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(107, 108)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽¹⁰⁹⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁹²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
438 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITH MCC	n/a	4.7	CMS GMLOS
439 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITH CC	n/a	3.1	CMS GMLOS
440 DISORDERS OF PANCREAS EXCEPT MALIGNANCY WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
PANCREATITIS	13%	3.6	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List
- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Expected progress after 24-48h

- Pain decreasing and manageable with medications
- Vital signs within acceptable limits
- Fluid and electrolyte imbalances and malnutrition returning to acceptable limits
- Labs returning to acceptable limits (e.g., lipase, amylase)
- Oral intake improving

Consider

- Increasing activity
- Nutritional consult for inadequate oral intake, low fat diet, and other diet information upon discharge home
- Discharge planning

4:

Care Facilitation

- Enteral feeding expected, arrange home nutrition
- Home nutrition therapy should be considered in a clinically stable patient who continues to require supplemental nutrition due to the inability to tolerate a solid, low-fat diet (**Home**)

Potential risk of readmission, consider an alternate level of care or readmission reduction program

- Patient who have an inability to tolerate a solid diet, gastrointestinal symptoms (nausea, vomiting, and diarrhea), or uncontrolled pain (**Home Care, Readmission reduction program**)
- Knowledge deficit or non-adherence with condition management identified (**Home Care, Readmission reduction program**)
- Acute pancreatitis admission within the last 6 months (**Home Care, Readmission reduction team**)
- Co-morbid chronic obstructive pulmonary disease, diabetes, chronic kidney disease, atrial fibrillation (**Home Care, Readmission reduction team**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs,

home homemaking services, transportation resources

- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., *Pediatr Rev* 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). *Psychosis and schizophrenia in adults: prevention and management*. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow up (Centers for Disease Control and Prevention, *Lead Poisoning Prevention*. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., *CMAJ* 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., *PLoS One* 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination

of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

Abdominal pain characteristic of pancreatitis is usually located in the epigastric region and may radiate to the back. The onset is often abrupt, progressive and persistent. The pain can often be associated with nausea and vomiting (Waller et al., J Emerg Med 2018, 55: 769-79; Banks et al., Gut 2013, 62: 102-11; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16). Over 80% of patients with chronic pancreatitis will have acute abdominal pain that is associated with recurrent pancreatitis (Singh et al., JAMA 2019, 322: 2422-34).

7:

Although both amylase and lipase are elevated in acute pancreatitis, lipase is more specific for pancreatitis (Abu-El-Haija et al., J Pediatr Gastroenterol Nutr 2018, 66: 159-76; Waller et al., J Emerg Med 2018, 55: 769-79; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16). Amylase and lipase levels may be within normal limits in patients with CP due to loss of pancreatic exocrine function due to fibrosis (Lew et al., Dig Dis Sci 2017, 62: 1702-12).

8:

Guidelines recommend that a baseline abdominal ultrasound (US) be performed upon admission for acute pancreatitis. Transabdominal ultrasound (US) is the preferred method to detect cholelithiasis and bile duct dilatation validating its use in confirming these etiologies in pancreatitis (Greenberg et al., Can J Surg 2016, 59: 128-40; Banks et al., Gut 2013, 62: 102-11). Ultrasound findings that are consistent with acute pancreatitis include (Banks and Freeman, Am J Gastroenterol 2006; 101(10): 2379-2400):

- Diffuse glandular enlargement
- Hypoechoic texture of the pancreas reflective of edema
- Ascites

9:

Imaging in pancreatitis may consist of:

- Abdominal ultrasound on admission
- Contrast-enhanced computed tomography
- Magnetic resonance imaging

10:

Contrast-enhanced computed tomography (CT) of the abdomen may be performed in specific circumstances for acute pancreatitis. CT with contrast is most useful after 48-72 hours after admission, as findings can be negative if performed too early. MRI is used in circumstances that require differentiating soft tissue pathology and evaluation of biliary and pancreatic ductal obstruction. Examples of reasons to perform a CT with contrast or MRI include (Waller et al., J Emerg Med 2018, 55: 769-79; Greenberg et al., Can J Surg 2016, 59: 128-40):

- When the diagnosis is uncertain
- To determine severity
- Identifying peripancreatic fluid collections
- Estimating degree of pancreatic necrosis
- When Systemic Inflammatory Response Syndrome (SIRS) is persistent
- For continued clinical deterioration despite traditional treatment after 72 hours of illness
- To exclude alternative intra-abdominal processes
- To identify local complications of pancreatitis, such as peritonitis

CT or magnetic resonance (MRI) imaging are first line for the diagnosis of chronic pancreatitis. Endoscopic ultrasonography (EUS) may be indicated in chronic pancreatitis if the diagnosis remains in question, but it is invasive and lacks specificity. If diagnosis is not confirmed by CT, MRI, or EUS, secretin-enhanced magnetic resonance cholangiopancreatography may be indicated if clinical suspicion is high (Gardner et al., Am J Gastroenterol 2020, 115: 322-39).

11:

Computed tomography (CT) findings consistent with acute pancreatitis include (Banks et al., Gut 2013, 62: 102-11):

- Enlargement of the pancreas with diffuse edema
- Heterogeneity of pancreas parenchyma
- Peripancreatic stranding
- Peripancreatic fluid collections

Key findings on CT imaging for chronic pancreatitis are (Conwell et al., Pancreas 2014, 43: 1143-62):

- Parenchymal loss
- Chronic inflammation
- Fibrosis

12:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

13:

Risk factors for severe acute pancreatitis include (Ham et al., Emerg Med Clin North Am 2017, 35: 571-87; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16):

- Patient characteristics (such as age > 55 years, comorbid health problems, altered mental status, and BMI > 30)
- Presence of SIRS
- Laboratory findings consistent with hypovolemia:

- Blood urea nitrogen > 20 mg/dl or rising

- Hematocrit > 44% or rising

- Radiology findings of pleural effusions or infiltrates

14:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

15:

Gallstones in the common bile duct (choledocholithiasis) occur in 10-20% of patients with cholelithiasis (gallstones). Gallstone pancreatitis is a serious complication of choledocholithiasis associated with increased morbidity and mortality. Patients may present with jaundice, RUQ pain, nausea, vomiting, and elevated bilirubin level. Strong predictors of choledocholithiasis include evidence of common bile duct stone on ultrasound, presence of acute cholangitis, and a serum bilirubin greater than 4 mg/dl. Treatment includes clearing the obstructing stones by endoscopic retrograde cholangiopancreatography (ERCP) or surgery (Wilkins et al., Prim Care 2017, 44: 575-97).

16:

Pleural effusion has been reported in up to 50% of patients with acute pancreatitis as detected on computerized tomography. Findings include mild to moderate effusions, usually on the left side. Chest radiograph is the primary imaging for pleural effusion; however, chest computerized tomography is more sensitive in detecting minimal amounts of pleural fluid. Ultrasound at the bedside is the preferred method in the intensive care setting. Pleural effusions usually resolve as inflammation subsides. Effusions that last longer than 2 weeks, or are severe, may be associated with pancreatic pseudocyst or pancreaticopleural fistula and may require drainage, but these occurrences are rare (Kumar et al., JGH Open 2019, 3: 71-9).

17:

Intrathoracic complications have been attributed to 22-29% of death in patients with pancreatitis. Common

pulmonary complications include atelectasis, pneumonia, acute respiratory distress syndrome. Pneumonia may be secondary to sepsis, infected pancreatic necrosis, malnourishment, or mechanical ventilation. Imaging findings of pneumonia associated with pancreatitis are similar to pneumonia in other conditions (Kumar et al., JGH Open 2019, 3: 71-9).

18:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

19:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels. PCO_2 values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

20:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

21:

Instruction: Room air assessment may not be clinically appropriate in patients who are dyspneic, tachypneic, cyanotic, or experiencing moderate to severe respiratory distress. If a patient remains hypoxic while on oxygen, room air assessment is not required.

22:

Systemic inflammatory response syndrome (SIRS) is an inflammatory process defined as 2 or more of the following clinical findings: fever, leukocytosis, tachypnea, and tachycardia.

23:

The presence of systemic inflammatory response syndrome (SIRS) in the initial 24h is highly sensitive for predicting organ failure, and persistent SIRS is associated with multiorgan dysfunction and severe acute pancreatitis. Patients may require re-evaluation of the severity of pancreatitis daily as symptoms are evolving (Banks et al., Gut 2013, 62: 102-11).

24:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

25:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with

outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

26:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

27:

In patients hospitalized with pancreatitis, mortality within the first two weeks of hospitalization is usually the result of multisystem organ failure. Organ failure is a better predictor of morbidity and mortality than is the extent of pancreatic necrosis (Anand et al., Gastroenterol Clin North Am 2012, 41: 1-8). Many patients presenting with acute pancreatitis may not have evidence of organ failure on admission, and it is important to identify the patients who may require more aggressive treatment and monitoring. The Revised Atlanta Criteria recommends using a score of 2 for one of the respiratory, cardiovascular, or renal systems using the modified Marshall scoring system to identify organ failure. In lieu of the modified Marshall score, continued use of the Atlanta definition of organ failure may be used (Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16):

- Shock (SBP 90 mm Hg)
- Pulmonary insufficiency (PaO₂ 60%)
- Renal failure (creatinine > 2 mg/dl after rehydration)
- GI bleeding (> 500 cc/24 h)

28:

Patients with severe pancreatic disease should be treated at the critical level of care to facilitate early recognition and management of complications. Clinical scoring systems to accurately determine severity are difficult to use, especially in the first 24 hours of admission (Waller et al., J Emerg Med 2018, 55: 769-79).

29:

Abnormalities of calcium can have significant impact on morbidity and mortality due to its crucial role in neuromuscular and cardiac function. Patients who present with symptoms usually have an abrupt decline in ionized calcium.

Common symptoms of hypocalcemia include:

- Perioral numbness
- Carpopedal spasm of the hands or feet
- Flaccid paralysis
- Muscle weakness
- Paresthesia
- Tetany

Symptomatic hypocalcemia requires treatment with intravenous calcium (Turner et al., Endocr Connect 2019, 8: X1).

30:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma;

FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

31:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

32:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552). Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

33:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

34:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

35:

Symptoms of hypernatremia include:

- hyperreflexia
- irritability
- mental status changes or Glasgow Coma Scale (GCS) 9-14
- muscle weakness
- restlessness

36:

Antibiotics should not be routinely administered to prevent infection. Studies have shown they are not beneficial, do not prevent morbidity or mortality, and increase the risk of developing fungal infections. Treatment with antibiotics is appropriate if there is a clinical suspicion of infection, or after confirmation of infection via blood cultures (Greenberg et al., Can J Surg 2016, 59: 128-40; Dick et al., J Hosp Med 2016; Goldenberg et al., Expert Rev Gastroenterol Hepatol 2014, 8: 687-94).

37:

In a large cohort study of over 4,000 patients with acute pancreatitis pain, about 80%(0.80) received opioids during the initial stages of hospitalization. Increased opioid use was found to be associated with a longer length of hospitalization, with about 10%(0.10) of patients continuing to use opioids after discharge. This continued use of opioids may reflect in chronic pancreatitis management (Wu et al., JAMA Netw Open 2019, 2: e191827).

38:

Aggressive hydration in patients with acute pancreatitis is recommended in the first 12-24h of care and frequent assessment for the first 24-48h of admission (Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16). Goal-directed fluid therapy is recommended to prevent complications associated with over hydration, such as respiratory complications and abdominal compartment syndrome (Crockett et al., Gastroenterology 2018, 154: 1096-101). Frequent re-evaluation, at least every 6 hours, is suggested with a goal to decrease blood urea nitrogen levels and produce a urine output of at least 0.5 ml/kg/h. A total volume of fluid in the first 24h of 2.5-4L is the goal with adjustments for patients age, weight, physical exam, and comorbid conditions (Janisch and Gardner, Gastroenterol Clin North Am 2016, 45: 1-8).

39:

Fluid resuscitation in severe dehydration involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. The administration of 20 mL/kg boluses until adequate urine output, resolution of symptoms, or adequate hydration without further losses is recommended (Piescik-Lech et al., Aliment Pharmacol Ther 2013, 37: 289-303).

Patients who received IV fluid totaling 2.5 L per day or more require close electrolyte monitoring as this amount can cause significant hyponatremia. For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used. For patients with obesity, adjust IV fluid rates to their ideal body weight (National Institute for Health and Care Excellence, Intravenous fluid therapy in adults in hospital. 2017).

40:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

41:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

42:

Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection. Patients presenting with sepsis are associated with hospital mortality rates of approximately 10%(0.10). Septic shock is a subset of sepsis that is associated with cardiovascular dysfunction and cellular/metabolic abnormalities so profound, mortality rates are greater than 40%(0.40). Both sepsis with organ dysfunction and septic shock should be managed at the Critical level of care (Shankar-Hari et al., JAMA 2016, 315: 775-87; Singer et al., JAMA 2016, 315: 801-10; Dellinger et al., Crit Care Med 2013, 41: 580-637).

43:

Acute cholangitis is an infection of the biliary tree and occurs as a consequence of biliary tract obstruction. The clinical presentation is variable from mild symptoms of fever, abdominal pain, and jaundice, to severe sepsis. Diagnosis is based on clinical presentation and imaging studies. Early CT or ultrasound is recommended to evaluate biliary dilatation and possible liver abscess. In patients with mild disease, treatment includes anti-infective therapy and supportive treatments such as analgesia and intravenous fluid. These patients should undergo biliary drainage within 24-48 hours. Patients with sepsis should be managed at the Critical level of care and often require fluid resuscitation and urgent biliary drainage which may be performed by endoscopic, surgical, or percutaneous approaches (Sartelli et al., World J Emerg Surg 2013, 8: 3; Attasaranya et al., Med Clin North Am 2008, 92: 925-60, x).

44:

Common findings for patients with biliary sepsis include:

- T > 99.4° F(37.4°C)
- WBC > 15,000 cu.mm(15x10⁹/L)

- Bilirubin > 2.5 mg/dL(42.8 µmol/L)
- Jaundice
- Evidence of obstruction on imaging

45:
Calcium levels within criteria represent total serum calcium.

46:
Hypernatremia (Seay et al., Am J Kidney Dis 2020, 75: 272-86)

47:
Hypovolemic patients should be resuscitated with 0.9% sodium chloride regardless of the serum sodium level until their vital signs are normal. Normovolemic patients with hypernatremia should receive either 0.45% sodium chloride or 5% dextrose infusion (Muhsin and Mount, Best Pract Res Clin Endocrinol Metab 2016, 30: 189-203).

48:
The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

49:
Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

50:
Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

51:
Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

52:
Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

53:
Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.
AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.
AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309;

Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

54:

In this instance, oliguria is not the result of an outlet obstruction.

55:

Urine output less than 0.5 mL/kg/h may be considered oliguria.

56:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

57:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Pistolesi et al., J Nephrol 2018, 31: 797-812; Joannidis et al., Intensive Care Med 2017, 43: 730-49; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

58:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

59:

Dialysis includes hemodialysis or peritoneal dialysis. Both modalities are equally effective and should be provided based on patient needs and device availability (Liu et al., Cochrane Database Syst Rev 2017, 12: Cd011457).

60:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

61:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

62:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

63:

Prior to the initiation of immunosuppressants for systemic diseases, which adversely impacts kidney function, a renal biopsy may be necessary.

64:

The American Gastroenterological Association Institute Guideline on Initial Management of Acute Pancreatitis

recommends against routine use of urgent ERCP (Crockett et al., Gastroenterology 2018, 154: 1096-101). However, in patients with acute pancreatitis and cholangitis or choledocholithiasis with common bile duct dilatation found on ultrasound, an ERCP should be performed within 24-72h of admission (Waller et al., J Emerg Med 2018, 55: 769-79; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16).

65:

Endoscopic retrograde cholangiopancreatography (ERCP) is commonly used to treat pain and manage local complications in patients with chronic pancreatitis; particularly in patients who have pain that is caused by an obstruction to the main pancreatic duct due to strictures or stones (Valente et al., Scand J Surg 2020, 109: 59-68; Drewes et al., Pancreatology 2017, 17: 720-31). In addition, ERCP may be used in patients who fail clearance of stones after using adequate Extracorporeal Shock Wave Lithotripsy (Hart and Conwell, Am J Gastroenterol 2020, 115: 49-55; Schepers et al., Lancet 2020, 396: 167-76).

66:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

67:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

68:

This criteria point is satisfied by any DVT:

- Lower extremity
- Upper extremity
- Splanchnic vein

69:

Thrombosis of the splanchnic veins (including splenic, superior mesenteric, and portal veins) is associated with severe, acute pancreatitis in more than one fifth of cases (Norton et al., Hepatobiliary Pancreat Dis Int 2020, 19: 116-21). Patients diagnosed with necrotizing pancreatitis are at increased risk for venous thromboembolism including splanchnic vein thrombosis, and less frequently extremity deep vein thrombosis and/or pulmonary embolism (Maatman et al., J Am Coll Surg 2021, 232: 91-100). Risk factors for venous thromboembolism in patients with necrotizing pancreatitis include organ failure (respiratory or cardiac) and infected necrosis. Further research is needed to determine the association of necrosis treatment and development of splanchnic vein thrombosis, as well as the utility of anticoagulation therapy.

70:

Although deep vein thrombosis may be suspected with symptoms of swelling, pain, and warmth, it can definitively be diagnosed by ultrasound (Desjardins et al., J Am Coll Radiol 2020, 17: S315-s22; Hanley et al., J Am Coll Radiol 2018, 15: S413-s7). Ultrasound is an easily accessible diagnostic tool that can be performed at the bedside, allowing for prompt diagnosis (Tritschler et al., Jama 2018, 320: 1583-94).

71:

Anticoagulation may include the following:

- Direct oral anticoagulants (DOACs)
- Low molecular weight heparin (LMWH) (lovenox, arixtra, fragmin)

- Unfractionated heparin
- Fondaparinux
- Argatroban
- Bivalirudin

72:

While low molecular weight heparin (LMWH) generally is considered first-line treatment for patients with symptomatic thromboembolic disease when anticoagulation is indicated, there are specific clinical circumstances in which unfractionated heparin is preferable, including (Stevens et al., Chest 2021, 160: e545-e608; Ortel et al., Blood Adv 2020, 4: 4693-738):

- Renal failure (CrCl < 30 mL/min)
- Morbid obesity (BMI ≥ 40 kg/m²) or weight > 120 kg
- Antiphospholipid antibody syndrome
- Bleeding

Risk of bleeding, including, but not limited to:

- Anticipated surgical intervention or interventional procedure (e.g., hemodynamic instability, thrombolysis)
- Anticipated need for urgent discontinuation

73:

An ileus is a blockage, partial or complete, or paralysis of the bowel that prevents the passage of intestinal contents from moving forward. It is more common in the small bowel, as opposed to large bowel. It is often the result of surgical adhesions or hernia. Small bowel ileus symptoms typically present as acute and severe, with complaints of nausea and vomiting, cramps, bloating, and the retention of stool and flatus. It can often be treated medically. Large bowel ileus usually has a milder presentation and often is due to cancer. It may also be the result of adhesions or stenoses from recurrent diverticulitis or, less commonly, is associated with a sigmoid volvulus or hernia (Vilz et al., Dtsch Arztebl Int 2017, 114: 508-18).

74:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

75:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

76:

Initiation of enteral nutrition refers to the introduction of fluid into the gastrointestinal tract via oral intake or gastrointestinal tube feeding (e.g., G-tube, J-tube).

77:

Guidelines for acute pancreatitis recommend starting oral feedings within 24 hours of admission. If unable to tolerate oral feedings, enteral tube feeding is preferred over parental nutrition. Parenteral nutrition should be reserved for those patients unable to tolerate enteral feeds due to the associated risk of organ failure (Crockett et al., Gastroenterology 2018, 154: 1096-101; Greenberg et al., Can J Surg 2016, 59: 128-40; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16). In chronic pancreatitis, there may be resulting maldigestion, secondary malabsorption, and nutrient deficiencies. Supportive nutritional therapy for these patients often includes pancreatic enzyme replacement therapy (PERT). For patients unable to tolerate an oral diet, enteral nutrition is recommended via the best tolerated route along with peptide semi-elemental products to optimize absorption. In cases of chronic pancreatitis with severe malabsorption, parenteral nutrition may be indicated (Dominguez-Munoz

and Phillips, Gastroenterol Clin North Am 2018, 47: 95-106;Maev et al., Ter Arkh 2016, 88: 81-9).

78:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

79:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

80:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

81:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

82:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

83:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level

services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

84:

Expected progress after 48-72h

- Pain decreasing and manageable with medications; dose and/or frequency of medications decreasing
- Afebrile with stable vital signs and within acceptable limits
- Fluid and electrolyte imbalances and malnutrition resolving and returning to acceptable limits
- Labs returning to acceptable limits (e.g., lipase, amylase)
- PO medications and adequate oral nutrition
- Activities at or close to baseline
- Discharge anticipated within the next 24 hours

If no, consider

- Nutritional consult for patients with inadequate oral intake
- Specialist consult, such as general surgery or infectious disease, for patients who are demonstrating infection and pancreatic necrosis
- Pancreatic imaging for persistent fever or pain

85:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

86:

The anticoagulation regimen may include low molecular weight heparin (LMWH), unfractionated heparin (UFH), or fondaparinux with or without warfarin, or warfarin alone. For adult patients, non-vitamin K oral anticoagulants (NOACs) (also known as novel oral anticoagulant, direct oral anticoagulant (DOAC), and target-specific oral anticoagulant) such as dabigatran, rivaroxaban, apixaban, or edoxaban, may also be used (Ortel et al., Blood Adv 2020, 4: 4693-738).

87:

Home therapy with low molecular weight heparin (LMWH) has been shown to be as safe as inpatient administration of LMWH or unfractionated heparin for patients with uncomplicated deep vein thrombosis (DVT) (Ortel et al., Blood Adv 2020, 4: 4693-738).

88:

The American Gastroenterological Association Institute Guideline on Initial Management of Acute Pancreatitis recommends against routine use of urgent ERCP (Crockett et al., Gastroenterology 2018, 154: 1096-101). However, in patients with acute pancreatitis and cholangitis or choledocholithiasis with common bile duct dilatation found on ultrasound, an ERCP should be performed within 24-72h of admission (Waller et al., J Emerg Med 2018, 55: 769-79; Tenner et al., Am J Gastroenterol 2013, 108: 1400-15; 16).

89:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

90:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

91:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

92:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

93:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

94:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

95:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

96:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

97:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

98:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit

- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

99:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

100:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

101:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

102:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

103:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

104:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

105:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

106:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

107:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

108:

Prolonged mechanical ventilation (PMV) has been defined in the literature using a variety of time frames ranging from 2 to more than 29 days of mechanical ventilation required. The most widely accepted definition of PMV is from the Centers for Medicare and Medicaid Services (CMS) and is defined as 21 days of mechanical ventilation for at least 6 hours per day (Rose et al., Cochrane Database Syst Rev 2014, 6: CD009235). The most notable causes for PMV include medical illnesses, predominately pneumonia, exacerbation of chronic obstructive pulmonary disease (COPD), and complications following cardiac surgical procedures (Scheinhorn et al., Chest 2007, 131: 76-84; MacIntyre et al., Chest 2005, 128: 3937-54).

109:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.